

Robust Vectorized Surface Reconstruction with 2D-3D Joint Optimization

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ABSTRACT

In this work, we propose a robust pipeline to create vectorized models from LiDAR point clouds without the assumption of watertight polygonal surfaces. The core idea behind our method is combining the hierarchical 2D projection with the pairwise intersections of clipped planes to precisely hypothesize faces of an object inside and outside while enhancing the robustness of different scenes. Moreover, to generate concise polygon meshes efficiently, Adaptive K-means++ (AKM++) is used to smooth the 2D rasterized height map followed by the Levenberg-Marquardt (LM) algorithm to optimize the projection primitive with planar intersections jointly. The final model could be easily obtained by merging the derived neighbouring facets above. Our experimental results show that the proposed method obtains compact models with high fidelity and efficiency on either precise or defect-laden data compared with other state-of-the-art approaches.

CCS CONCEPTS

• Computing methodologies; • Mesh models;

KEYWORDS

Surface reconstruction, vectorized, watertight

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1 INTRODUCTION

The reconstruction of 3D vectorized models has attracted increasing attention due to its great potential for navigation, virtual reality and so on. Over the past few years, many methods have been

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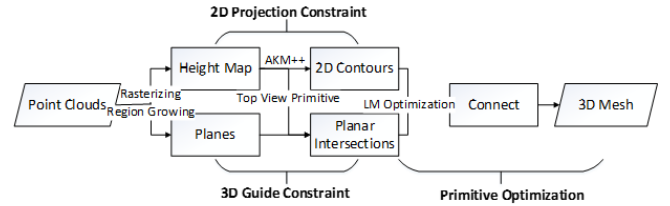


Figure 1: Core pipeline of our approach.

developed to create compact models, such as planar intersecting-based methods [Nan et al. 2017; Bauchet et al. 2020] and multistep 2D optimizing-based methods [Han et al., 2021]. However, almost all of these approaches enforce the watertightness as an important assumption and thus become fragile and compute-intensive when dealing with degraded cloud points of complex scenes.

This work introduces an effective pipeline to combine the rasterized projection with planar intersections to jointly optimize the contour of top view primitive with Levenberg-Marquardt (LM) algorithm. Moreover, in order to guarantee high efficiency of the reconstruction, adaptive K-means++ (AKM++) is innovatively used to remove noisy facet and create a compact model. In the following section, we introduce our proposed method, mainly focusing on three parts where major technical challenges are solved.

2 PROPOSED METHOD

The key observation that enables this design is that 2D contours constrain the basic shape of each facet while planar intersections guarantee the boundary accuracy. Hence, we refine the projection primitive with 3D planar intersections to keep the fidelity of meshes in different scenes (see Figure 1). In what follows, we introduce in detail three key parts of our method.

2D Projection Constraint. The rasterized height map can always display the sharp edge feature even in open-plan layout without the watertightness where the planar intersecting fails. Hence, we can easily extract the top view geometric facets from 3D points. In order to get the mask of each object robustly, we adaptively classify the map according to the height values and pixel distances with K-means++ [Arthur et al. 2006] into different clusters until the distance is below a threshold between centers. Compared with traditional planar detection with RANSAC, it can better filter noisy facets with small fluctuation partly. Another new idea to reduce the effect of defective point clouds is to cut off the binary contours

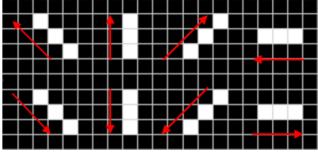


Figure 2: Eight principal orientations in rasterized contours.

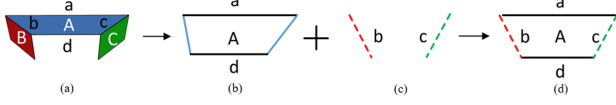


Figure 3: Whole steps of Part 3 (a) An open-plan layout consisting of primitive A, B, C; (b) The top view projection with individual edge a, d; (c) The 3D constraint with intersections b, c of A-B, A-C; (d) The primitive A refined by the 2D constraint a, d and 3D constraint b, c.

with eight principal orientations, so the contaminated edge must be very short in these orientations compared with the main body due to the randomness of noise. (see Figure 2). We further divide each part into N subsections sequentially and fit the vectorized edges rapidly via Dynamic Programming (DP) as

$$\min_{a_{ij}, b_{ij}, c_{ij}} \left(\max_{k \in [1, N_{ij}]} RMSE(x_{ij_k}, a_{ij}, b_{ij}, c_{ij}) \right), \quad (1)$$

where i, j, k denote section, subsection and pixel ID, respectively; a, b, c, x are line parameters and pixel in each subsection.

3D Guided Constraint. As the 2D boundary precision may reduce due to the deficiency of points, we choose to use the planar intersections to refine it furthermore. For each projection primitive, the correlative planes can be selected by the average distances of K nearest points to it. We only preserve the intersections of planes whose distances are below a specific threshold (see Figure 3c).

Primitive Optimization. As the 2D projection contour and the 3D planar intersections both have their advantages and disadvantages, we innovatively joint these two constraints together to get a precise vectorized model. We project the candidate intersections to the top view primitive and search the nearest one j for each subsection i . As the similarity of two edges is decided by both the rotation and position, we define the error termed D which is the quadratic sum from point x to line l with weight termed λ consisting of angle θ and distance d . Hence, the coarse-to-fine problem can be transformed into the following global energy optimization problem:

$$\min_l \lambda (d_{ij}, \theta_{ij}) D(l, x_i) + (1 - \lambda (d_{ij}, \theta_{ij})) D(l, x_j), \quad (2)$$

where $\lambda(d_{ij}, \theta_{ij}) = \frac{(d_{ij} - d_{min})(\theta_{ij} - \theta_{min})}{(d_{max} - d_{min})(\theta_{max} - \theta_{min})}$. The optimization problem can be solved conveniently via LM algorithm. The whole procedure is shown in Figure 3. Then the neighbouring structures such as roofs and walls which share the same edge with the top view projection primitive can be easily connected by topological relationships. By repeating the hierarchical 2D projection from outside to inside, we could finally get a complete 3D model with different structures.

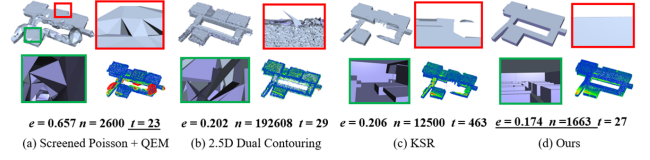


Figure 4: Reconstruction result on the MiMAP_02 benchmark [Wen et al., 2020; Wang et al., 2018] for Screened Poisson + QEM [Garland et al. 1997], 2.5D Dual Contouring [Zhou et al. 2012], KSR and Ours. Here e is the mean Hausdorff error (m) from sampled mesh points to input points; n is the number of facets of the final model; t is the running time (sec). The minimal e, n, t are underlined. The bottom right is the error map, where blue means the small value and red means the large one.

3 EXPERIMENTAL RESULTS

We perform an experiment that compares our proposed method with other state-of-the-art ones in both the public and private dataset. All the methods run in the same environment. Subject to the limited space, here we only show the result on MiMAP_02 benchmark, more results can be seen in our supplementary material.

In our experiment, we analyze the fidelity, efficiency and simplicity of each reconstruction approach. The result in Figure 4 shows that our pipeline has the advantage in both precision and simplicity although the running time is a little longer than that of Screened Poisson QEM. Besides, the red box indicates that our novel idea about AKM++ and analysis of eight principal orientations plays an important role in removing the defectives while smoothing the borders of each facet. The green box demonstrates the validity of our hierarchical 2D projection in recovering the complete structures both inside and outside. Moreover, our approach can adapt to different scenes, even in a layout without the watertightness.

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